

Weekend Work

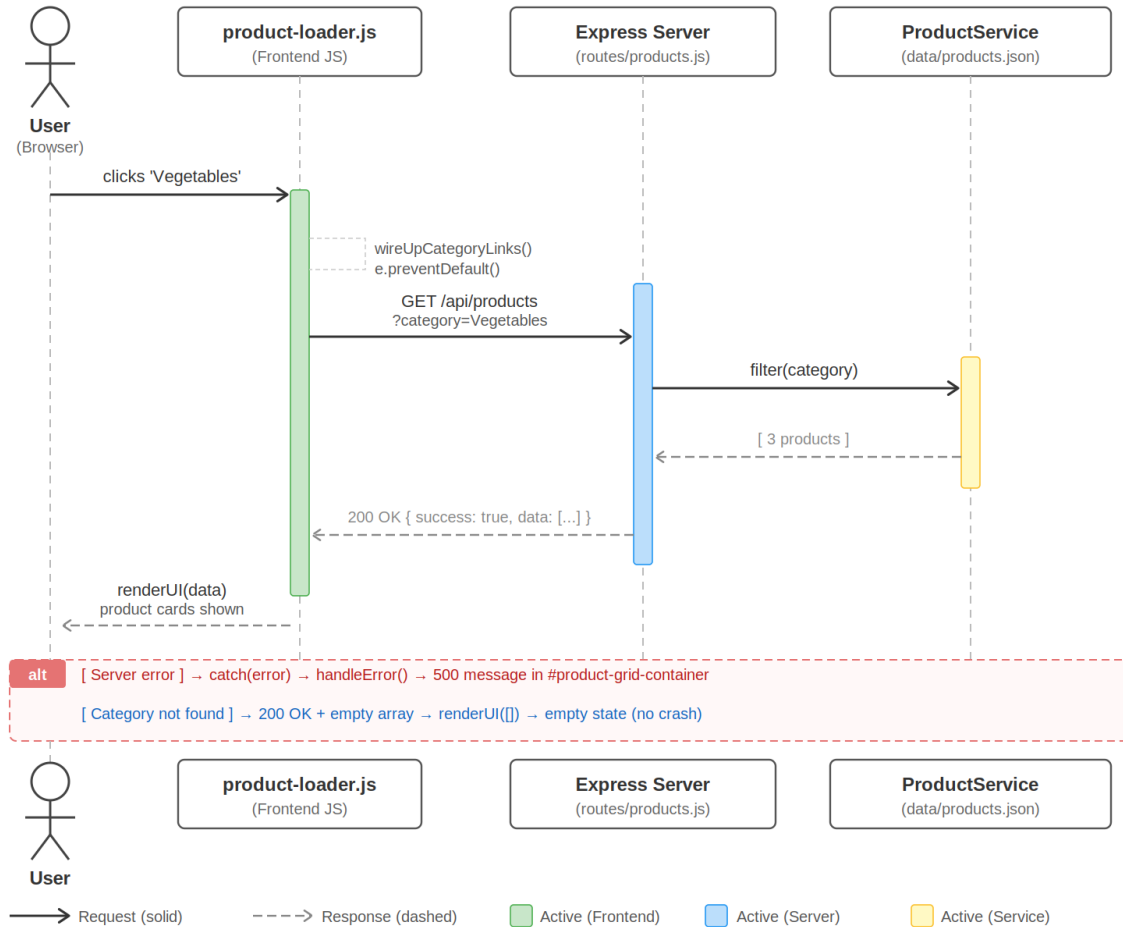
1. Write the “Contract” Table.

Component	Request (Order)	Response (Delivery)
Method	GET	-
Endpoint (URL)	/api/products?category={category}	-
Request Header	Accept: application/json	-
Response Header	-	Content-Type: application/json
Status Code	-	200 OK / 404 Not Found / 500 server Error
Body (Data)	(Empty – GET requests have no body)	{ “success”: true, “data”: [{ “id”: 1, “name”: “Fresh Organic Carrots”, “category”: “Vegetables”, “prices”: 4.50 }], “count”: 3 }

Example URLs:

- All products – GET /api/products
- Vegetables – GET /api/products?category=Vegetables
- Fruits – GET /api/products?category=Fruits

2. Write a Sequence Diagram for Request/Response Map.



3. Write GenAI Prompt from the “Contract” table + Sequence Diagram

- Express.js route for this request.
- Accept a query parameter for “category.”

Prompt:

"I have an organic grocery website called FreshmartOrganic built with HTML, CSS, and JavaScript.

Current Problem:

My file js/product-loader.js currently fetches product data from a static JSON file:

```
fetch('data/products.json')
```

I need to change it to call a REST API instead.

Backend Already Exists:

Express.js server running at localhost:3000 with these endpoints:

GET /api/products -> returns all products

GET /api/products?category=Vegetables -> returns only Vegetables products

Response shape: { success: true, data: [...products], count: N }

Each product has: id, name, category, price, image, status, discount

Available categories: Vegetables, Fruits, Dairy, Bakery, Natural, Grains

TASK — Rewrite js/product-loader.js so that:

1. On page load, call GET /api/products (not the static JSON file)
2. When user clicks a category link, call GET /api/products?category={name} Use e.preventDefault() to stop the page from reloading
3. Parse json.data from the response and pass it to renderUI()
4. If server returns error (500), show a user-friendly message inside #product-grid-container
5. Keep existing renderUI(), buildGridCard(), buildListCard(), cleanupLegacyProducts() — only change WHERE data comes from

Add inline comments for each function referencing this Logic Flow:

1. Trigger : User clicks category OR page loads
2. Request : Browser sends GET /api/products?category=xxx
3. Processing: Express server filters (Gatekeeper pattern)
4. Response : Server returns { success, data, count }
5. Render : renderUI(data) injects product cards into the DOM"